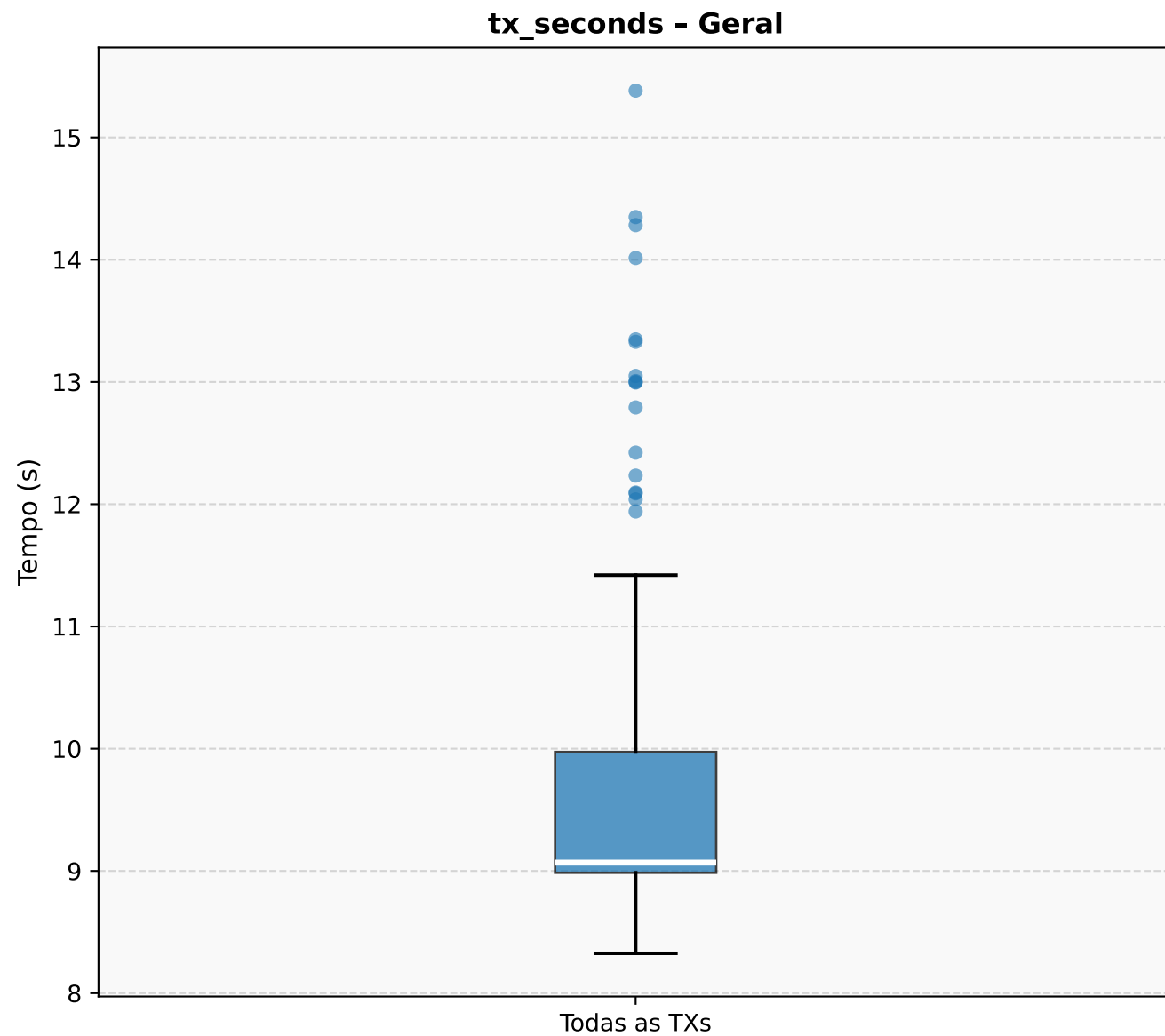
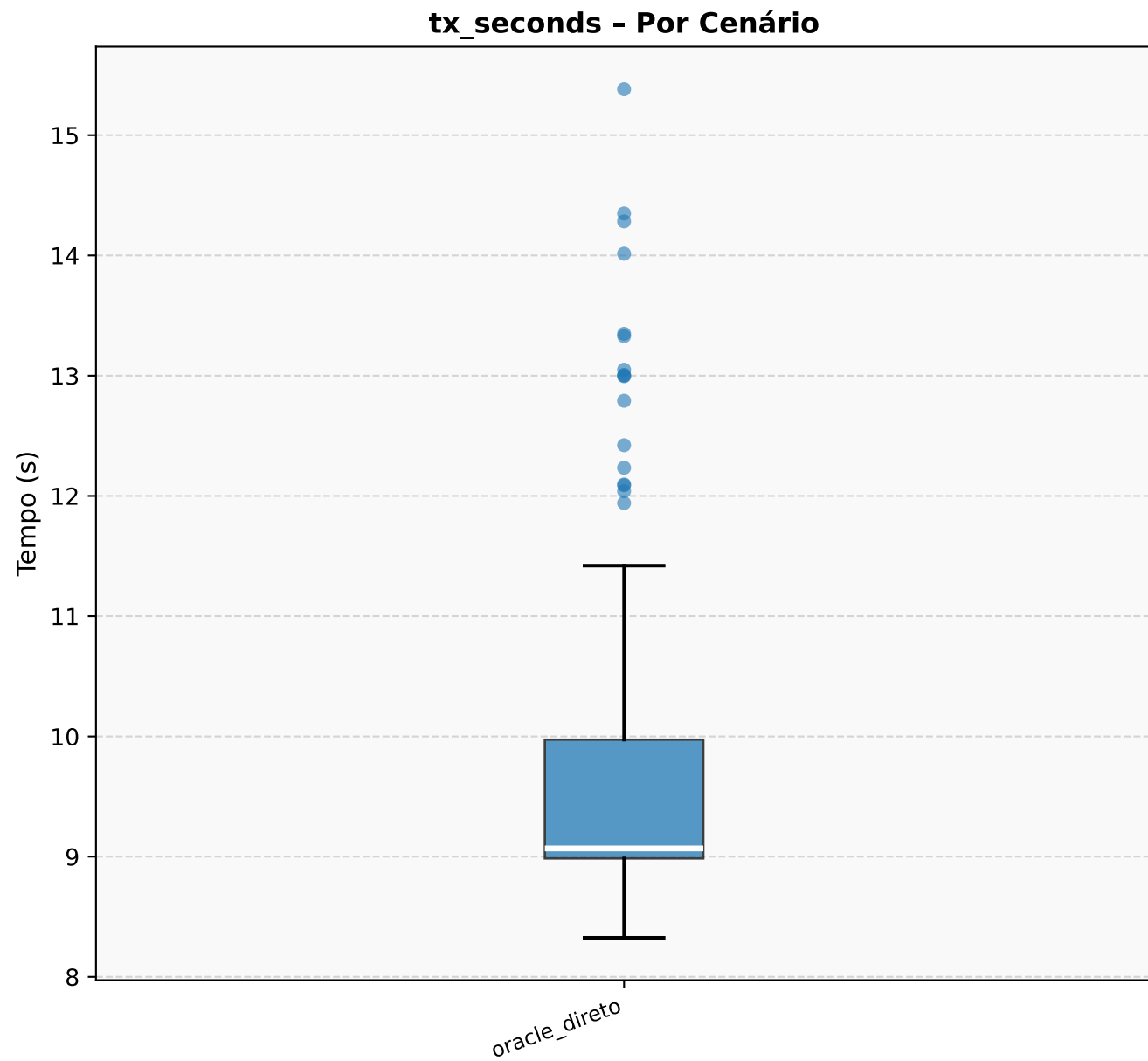


# Benchmark Hyperledger Besu - Tempo de Transação

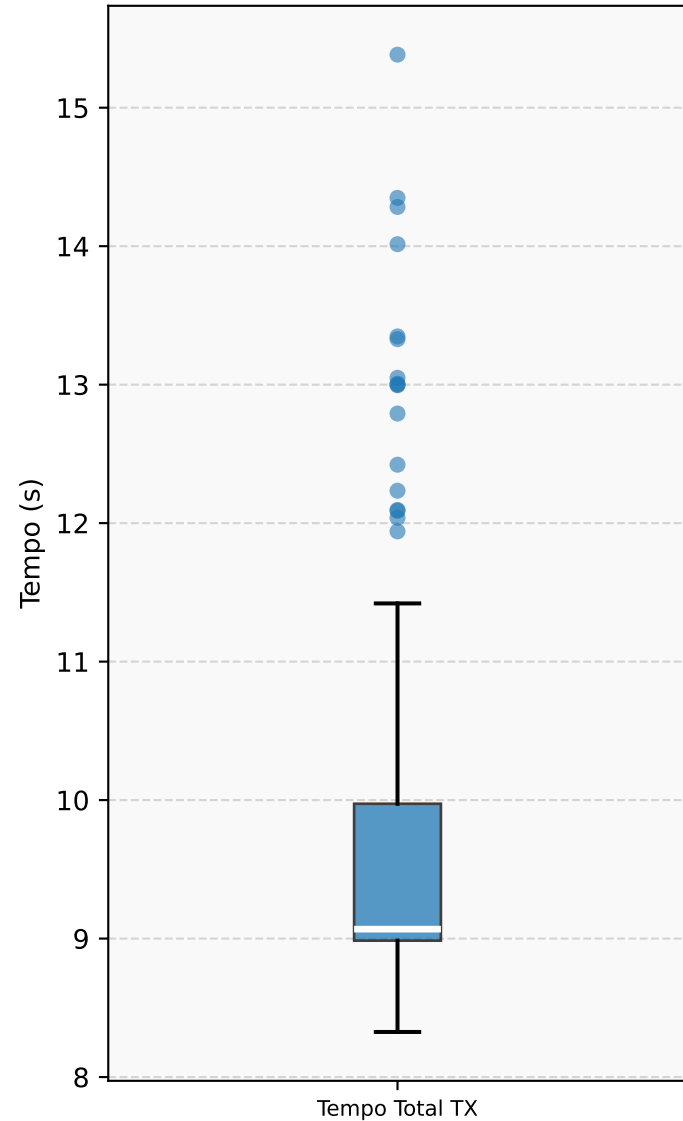


n=501 min=8.326s med=9.068s max=15.383s  $\sigma$ =0.915s



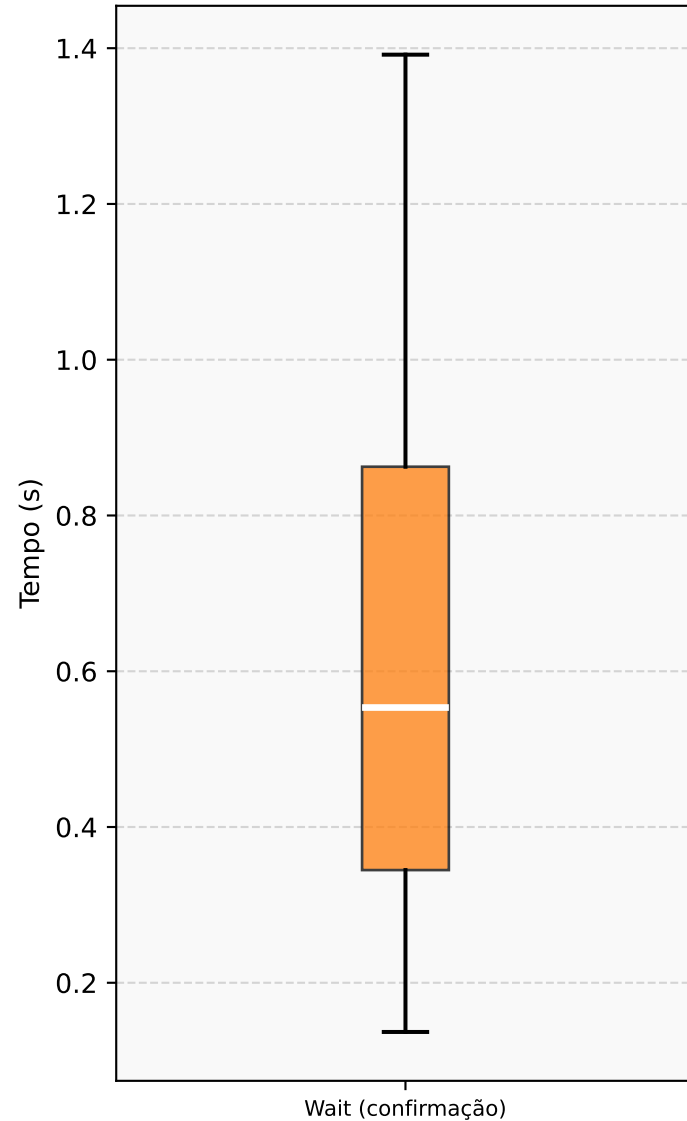
# Decomposição do Tempo por Componente

## Tempo Total TX



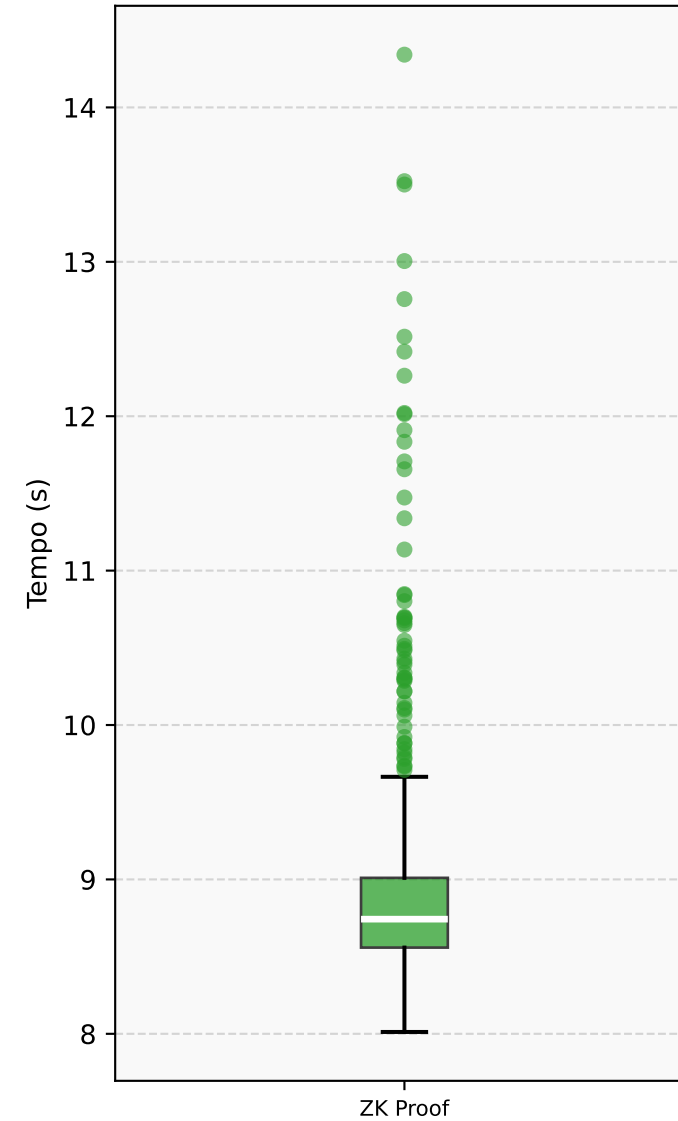
n=501 min=8.326s med=9.068s max=15.383s  $\sigma=0.915s$

## Wait (confirmação)



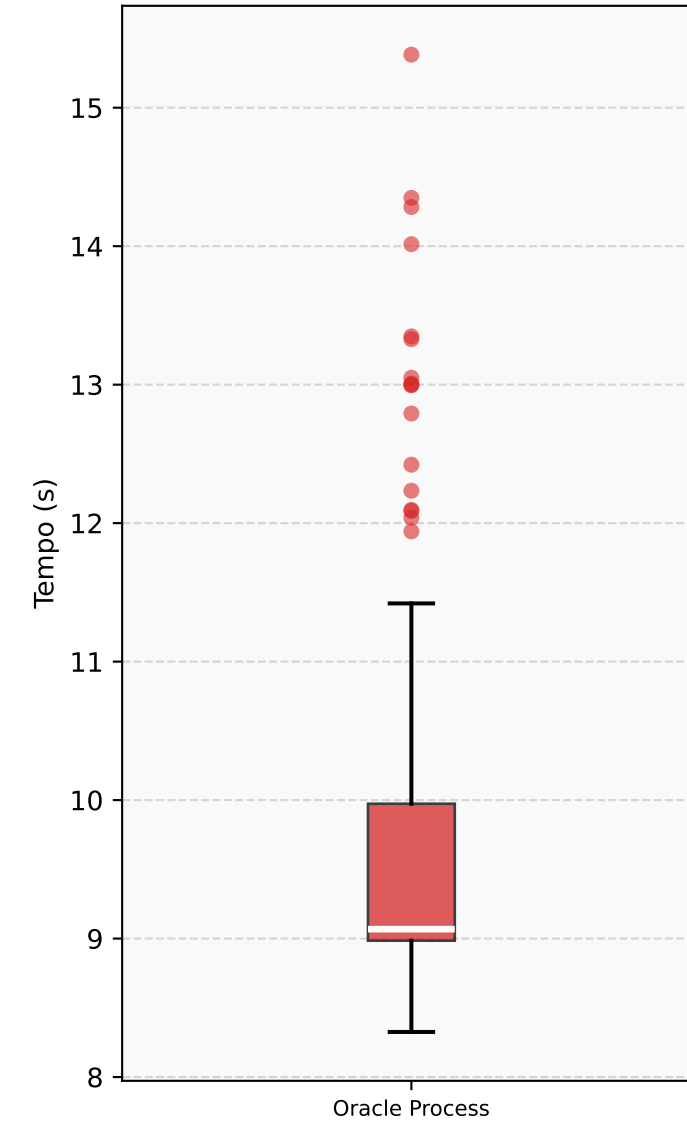
n=501 min=0.137s med=0.553s max=1.392s  $\sigma=0.319s$

## ZK Proof



n=501 min=8.012s med=8.743s max=14.342s  $\sigma=0.819s$

## Oracle Process



n=501 min=8.326s med=9.068s max=15.383s  $\sigma=0.915s$

tx\_seconds por Número de Bloco

